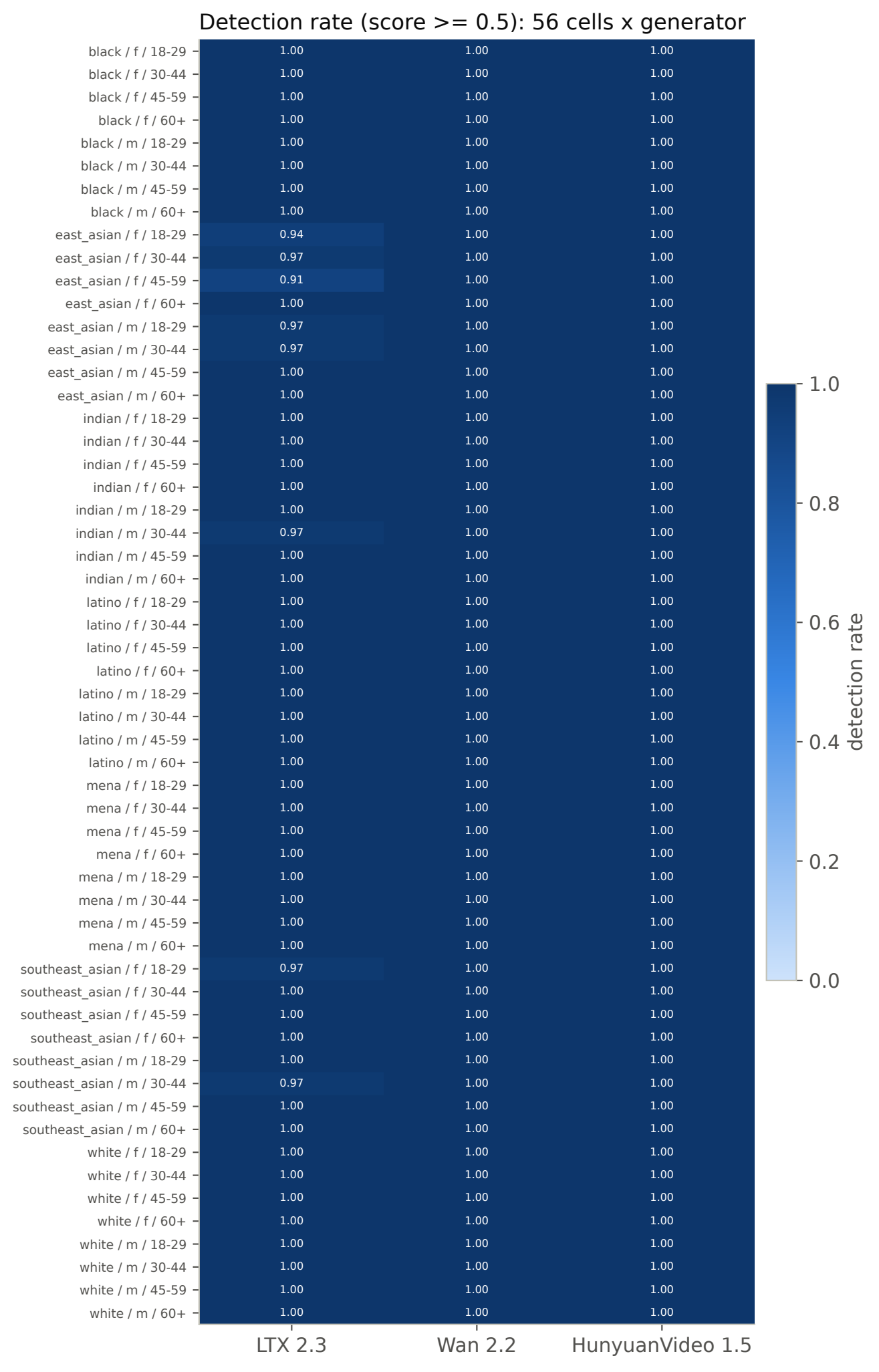

provisional threshold = 0.5 (detector default); replaced once reals exist